Finite residuals in the mixed zero fiber

Native audit · 42 fixed phi/doubling case-arms · 2026-10-25

Both full native processes pass all 42 arms each. Complete-tree parity, final source/execution seals and the terminal receipt pass, exit zero. No filter is deployed. Seven primes, three presentations per prime. Public scalar 271 and its presentations are not independent secret samples.

Native first: case-local totals	Phi	Doubling
Canonical S4 systems inspected	33,696	33,696
Eligible finite prime-field residuals	300	150
Retained non-prime-field residuals	33,396	33,546
Infinity residuals	0	0
Saved roots in eligible systems	65,532	32,766
Accepted representatives in eligible systems	0	0
Unique signed physical tuples before / after	504 / 504	504 / 504
Total historical F4 row multiplications	4,112,120,748	4,110,616,998
Historical F4 row multiplications in eligible systems	18,166,512	9,083,256

The eligible count is twice as large for phi. This is a counterfactual skip count, not a twofold speedup. The algebraic roots are saved symmetric triples, not physical relations. All eligible roots pass equation, component, zero-defect and fewer-than-three-distinct-cubic-root checks. No accepted tuple is lost. Each arm's exact physical set, augmented rows, zero-row count, independent rank and full-rank solution are preserved.



Figure 1: Literal first-execution counts and ranks for all 21 presentations in both arms. Rank preservation is within each arm; phi and doubling have different ranks on some curves. No time or full-cost gain is measured.

Why the predicate applies

Reject only a finite residual with x in Fp; retain infinity. The physical family has three distinct remaining ordinary abscissae, a fourth distinct peeled ordinary abscissa and one disjoint image. This does not authorize pruning an expanded signed-multiset relation family.

In $y^2 = x^3 + B$, the checked independence of $1,B,B^2,B^3$ makes all four component coefficients vanish when an S4 equation has four abscissae in Fp. Write a,b,c,d for the quartic elementary coordinates and $S = 64d - (4b-a^2)^2$. The existing identity $64S^3 = \sum A_i C_i$ forces $S = 0$ in the tested odd characteristics. The checked square bridge $512f(T) - 8Q(T)^2 = C_3T + 8S$, where $Q(T) = 8T^2 - 4aT + 4b - a^2$, then puts every quartic root on a quadratic with nonzero leading coefficient. Three distinct ordinary roots are impossible, including when the residual abscissa repeats one of them.

This reuses IDC1h65c11f01d0b1e67a and the prior ten-theorem Lean 4.34.1 gate: SquareIdeal.component_square_certificate, coefficients_force_defect_zero, components_exclude_three_distinct_roots and seven SquareRootBridge declarations. No new universal Lean theorem is claimed. Full field-embedding and group bridges remain unformalized. The fresh native program checks the actual field, points, residuals and saved roots. Infinity has a separate CM-triangle branch and stays retained.

The S4 necessity references are Gorla–Massierer, Definition 2.4 / Theorem 2.5 and Semaev, 2004/031. The prior proof report is under joint_image_elimination/reports/ run-20261024T0445Z-square/ in the parent study.

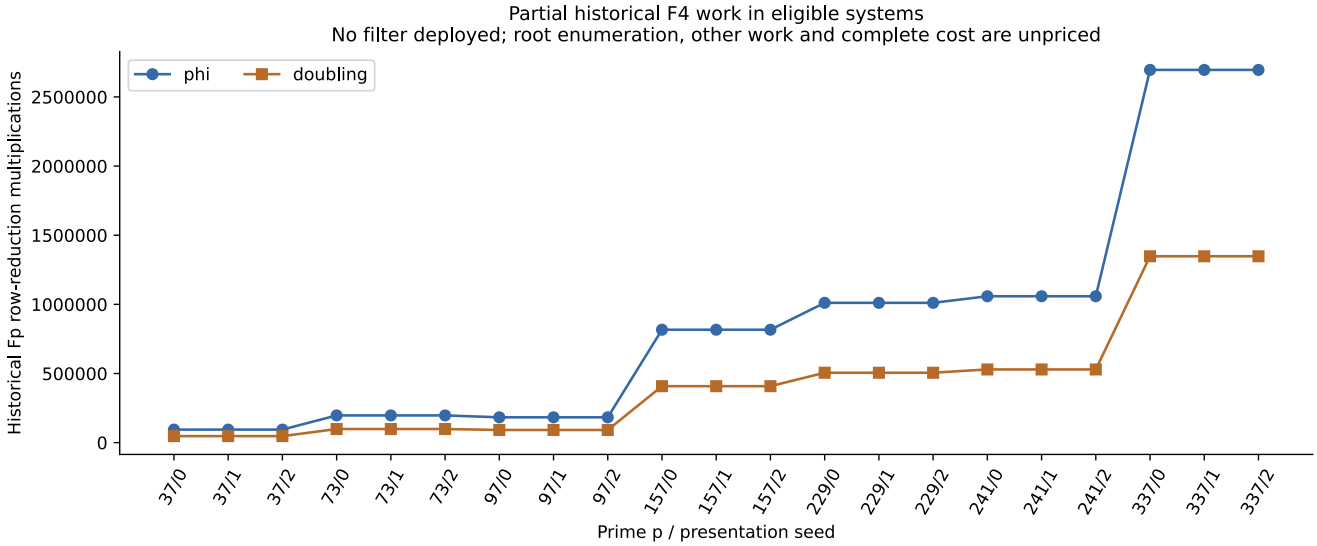


Figure 2: Partial historical row work in eligible systems. Root enumeration, construction, other F4 work, memory, hashing, allocation, I/O and the proposed predicate are unpriced. Many rejected roots do not price their enumeration. No actual filter or complete-cost advantage is established.

Verification and requirements retained

The independent Rust implementation rebuilds public orbit columns and 24 scalar actions with scalar multiplication, checks every physical five-point zero sum, transports retained representatives, compares exact sets and solves its own RREF. Full-rank factor logs are point checked. It does not call the old collector, equation deriver or RREF. The old complete field-root enumeration is reused; this is not a fresh exhaustive solver or an independent human review. Hash-bound missing or malformed input refuses execution; any mathematical counterexample is retained.

Eight synthetic controls pass with zero failures. Format, separate build and source seals pass. Source freeze before production: be9af2af1610. Binary SHA-256: d0f56b52bddfe3ef9a0731c4fdaeb65daa9e9d311e71a60e851406840005137f Workload SHA-256: 17ecb4c302526bc58e592466071e1c2704eaaf43a0a0ca1efe563e1bd69b3c10 Owner accepted 06:31:41 UTC, PID 3809049; first science admission 06:36:05 UTC. Each of the two processes keeps one 1,200-second/8 GiB phase under the common reservation. Queue wait is separate. Shared-host durations are diagnostic and unadmitted.

Requirement	Evidence / remaining gap
Original 189 + 21 cases	Complete; not repeated. No consistent twist-specific full cold advantage.
Unique relations / rank in 42-arm audit	Both full processes, complete-tree parity and terminal seals pass.
Original case-local ranks	p37 phi 2/2 vs doubling 1/2; p157 2/8 vs 1/8; p337 1/14 vs 3/14. Others zero. Each rank preserved by this rejection.
Public-only filtered collector	Not implemented here. Current results are counterfactual.
Complete cost from the twist	Open: complete cold collector/target work, calibration, strongest IC/rho and external controls.
85 target / 42 certificate / 12 degree configurations	Their original owners, inputs, budgets and full workloads are preserved.
Concrete q^2 instance proof	Two bounded memory failures; full failed evidence retained separately.

Next execute a public-only collector fork with the predicate charged before F4/root enumeration, infinity retained, and the same rule applied to doubling. Preserve all 42 exact physical sets and ranks and charge the whole cold pipeline and target recovery. A structurally rejected system may establish physical completeness; it must not report all algebraic roots enumerated or absent. Rank, complete price and reference gates remain.

Reproduction: zero_fiber_collector/fp_residual_audit/ contains REPORT.md, PROTOCOL.md, native/src/main.rs, build/run scripts and this source. Raw parent runs/: fp-residual-audit-native-gate-v1 and fp-residual-audit-v1. Per-case files retain every classification, root witness and both physical sets. The final archive is described in README.md.

Migration reconciliation at 06:35 UTC finds 28 terminal EKS jobs (18 complete, 10 failed, zero nonterminal), with no resubmission or role change. Diem stays expired at 50/52 and the legacy waiter stays quiesced. T-91 remains failed, attempt 4/4; no lease reset. Original receipt unchanged. No credentials enter the research artifacts. Canonical IC/rho graphs, leaderboard and benchmark index are unchanged because full-price and VALIDATION gates remain open. Report copies are unlisted.